



PART # 20-227-4 SN#: US3628-06
ENGINE: CRS XE275HL-10

GRIND#: CRS XE275HL-10
SPC INSTR 1:
SPC INSTR 2:

INTAKE EXHAUST
VALVE ADJUSTMENT .000 .000
GROSS VALVE LIFT .525 .525
DURATION @
.006 TAPPET LIFT 275 287
VALVE TIMING OPEN CLOSE
@ .006 INT: 32 BTDC 64
EXH: 78 BTDC 30

THESE SPECS ARE FOR CAM INSTALLED
@ 106.0 INTAKE CENTER LINE

INTAKE EXHAUST
DURATION @ .050 231 237
LOBE LIFT .3510 .3510
LOBE SEPARATION 110.0

ABDC
ATDC

995-16 SPRINGS REQUIRED
VALVE SPRING SPECS FURNISHED
WITH SPRINGS

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SERIAL#: US3628-06

**THIS CARD CONTAINS INFORMATION ABOUT
WARRANTY POLICIES.
PLEASE READ ALL THE INFORMATION AND
RETAIN THIS CARD FOR FUTURE REFERENCE.**

1. The attached specification card contains registration-specific information about your new COMP Cams® camshaft. Detach at perforation and keep for future reference.
2. Please take a minute to register your new COMP Cams® camshaft. Either fill out and return the attached warranty registration card or go online to www.compcams.com/warrantyregistration to submit the information electronically.
3. Refer to the enclosed instructions for proper installation and valuable tech tips. Instructions are also available online at www.compcams.com.
4. Call 1.800.999.0853 if you have further questions regarding component selection, installation or warranty information. Our technicians are there to help you with your valve train needs.

If you prefer, you can submit this information electronically at www.compcam.com/warrantyregistration

Name _____

Address _____

City _____ State _____ Zip _____

Phone () _____ Date Purchased _____

What product did you purchase from COMP Cams®? ☐ K-Kit ☐ SK-Kit ☐ CL-Kit ☐ RP-Kit ☐ Cam only

Who will install it? ☐ Yourself ☐ Engine Shop ☐ Garage ☐ Other

Why did you purchase this product from COMP Cams® rather than another camshaft company?

For a COMP Cams® full-line catalog and T-shirt, send \$15 with this card to
COMP Cams®.

Select size: M, L, XL, XXL, XXXL

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What type of vehicle will it be used in? ☐ Street Machine ☐ Street Rod ☐ RV ☐ Boat ☐ Street Truck
☐ 4x4 ☐ Muscle Car ☐ Oval Track ☐ Drag Race ☐ Daily Driver

What year model is your vehicle?

What is the single most important factor in choosing a camshaft company?

☐ Variety of grinds to choose from ☐ Availability (Cam in stock/available now rather than being ordered)

☐ Recommendation ☐ Advertisement ☐ Company Reputation ☐ Price

20-227-4

COMP

CRS XE275HL-10

SN#: US3628-06 06/05/06

CAM SPECS—INT EXH

ADV DUR: 275 287

DUR @ .050: 231 237 LOBE SEP: 110.0

VALVE LIFT: .525 .525



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11-28-05

Flat Tappet Camshafts

Recent changes in oil and engine technology are likely the cause of premature camshaft failure; here's what you can do to protect your engine!

Premature flat tappet camshaft failure has been an issue of late and not just with one brand or type of camshaft. In almost every case, the hardness or the taper of the cam lobe is suspected, yet most of the time that is not the problem. This growing trend is due to factors that are unrelated to camshaft manufacture or quality. Changes in today's oil products and "advanced" internal engine design have contributed to a harsher environment for the camshaft and a potential for failure during break-in. But there are several things you can do to turn the tide on this discouraging trend.

Proper Camshaft Set-Up & Break-In

Proper flat tappet camshaft set-up and break-in, as any engine builder knows, are keys to how long a camshaft will last, both short and long term. Making certain that the camshaft and lifters are properly lubricated will guarantee that the camshaft and lifters are protected during the critical initial start-up of your newly-built engine. COMP Cams® offers the right product for this job (Part #154), and it is available in several different size containers for engine builder convenience. To further enhance this "relationship," we strongly recommend the use of COMP Cams® Camshaft Break-In Oil Additive (Part #159) during the break-in. While this additive was originally developed specifically for break-in protection, subsequent testing has proven the durability benefits of its long term use. This special blend of additives promotes proper break-in and protects against premature cam and lifter failure by replacing some of the beneficial ingredients that the oil companies have been required to remove from off-the-shelf oil. These specialized COMP Cams® lubricants are the best "insurance policy" you can buy and the first step to avoiding durability problems with your new flat tappet camshaft.

Adequate Lubrication

Another major factor in the increase of flat tappet camshaft failure is your favorite brand of engine oil. Simply put, today's engine oil is just not the same as it used to be, thanks to ever tightening environmental regulations. The EPA has done a great job in reducing emissions and the effects of some of the ingredients found in traditional oils; however these changes to the oil have only made life tougher on your camshaft. The lubricity of the oil and specifically the reduction of important additives such as zinc and manganese, which help break-in and overall camshaft life, have been drastically reduced. In terms of oil selection, we recommend Shell Rotella T oil for the break-in procedure. Most often used in diesel engine applications, this higher lubricity oil works in gasoline engines as well.

Camshaft Installation and Degreeing Procedure

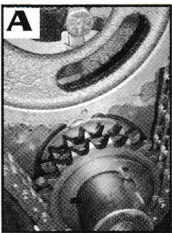
Thank you for choosing COMP Cams® products; we are proud to be your manufacturer of choice. Please read this instruction booklet carefully before beginning installation and also take a moment to review the included limited warranty information.

This instruction booklet is broken down into several categories, for ease of use. Some of the topics may not apply to every application, but all of the information will be very beneficial during the cam installation process. For step-by-step visual detail, it is recommended to watch the COMP Cams® video; "The Proper Procedure to Install and Degree a Camshaft" (Part #190-1). If there are any questions or problems during the installation, please do not hesitate to contact the toll free CAM HELP® line at 1-800-999-0853, 7 a.m. to 8 p.m. CST Monday through Friday.

Important: *In order for your new COMP Cams® camshaft to be covered under any warranty, you must use the recommended COMP Cams® lifters and valve springs. Failure to install new COMP Cams® lifters and valve springs with your new cam can cause the lobes to wear excessively and cause engine failure. If you have any questions about this application, please contact our technical department immediately.*

Camshaft Installation

Step 1: Prepare a clean work area and assemble the tools needed for the camshaft installation. It is suggested to use an automotive manual to help determine which items must be removed from the engine in order to expose the timing chain, lifters and camshaft. A good, complete automotive manual will save time and frustration during the installation.



Step 2: Once the camshaft, lifters, and timing chain are exposed, line up the timing marks on the timing gears by rotating the crankshaft (fig. A). This will position the #1 piston at Top Dead Center. Next remove the camshaft timing chain sprocket, the timing chain and the camshaft retaining plate if equipped. Remove all lifters and reinstall the cam sprocket to serve as a handle. Slowly and carefully "roll" the camshaft out of the engine. **Note:** *Excessive force is not required to remove a camshaft. If the camshaft does not come out easily, stop! Look for obstructions, such as a fuel pump rod, distributor gear, etc. Do not force the camshaft; something is holding it in the block.*

Step 3: Once the camshaft is removed, pull the bottom timing gear off of the crankshaft snout. In many cases you will need a gear puller to remove the crank sprocket. It usually has an interference fit, which may make it difficult to remove without the proper tool. Be careful not to damage the threads in the end of the crankshaft.

Step 4: Now is the time to inspect all of the old camshaft related components (timing chain set, distributor gear, etc.) for abnormal or excessive wear. **Note:** *Always use a new distributor gear with a new camshaft.* On flat tappet cams, it is critical that the old lifters be properly disposed of and new COMP Cams® lifters installed. Your camshaft warranty will be voided if old flat tappet lifters are used. With roller cams, roller lifters may be re-used. Inspect them thoroughly; look for any looseness in the wheels and excessive wear on the lifter body.

Step 5: Remove your new COMP Cams® camshaft from its packaging. Inspect all lobes and the distributor gear making sure the camshaft was not damaged in shipment. Next, compare the stamped numbers on the end of the cam with the spec card to make sure this is the correct cam, dowel pin length, journal spacing, distributor gear location. It is a good idea at this point to lightly wire brush the distributor gear with mineral spirits or an equivalent solvent. Remember, during the installation process, the cleaner you make your new components, the better chance you have of avoiding failure. Using